

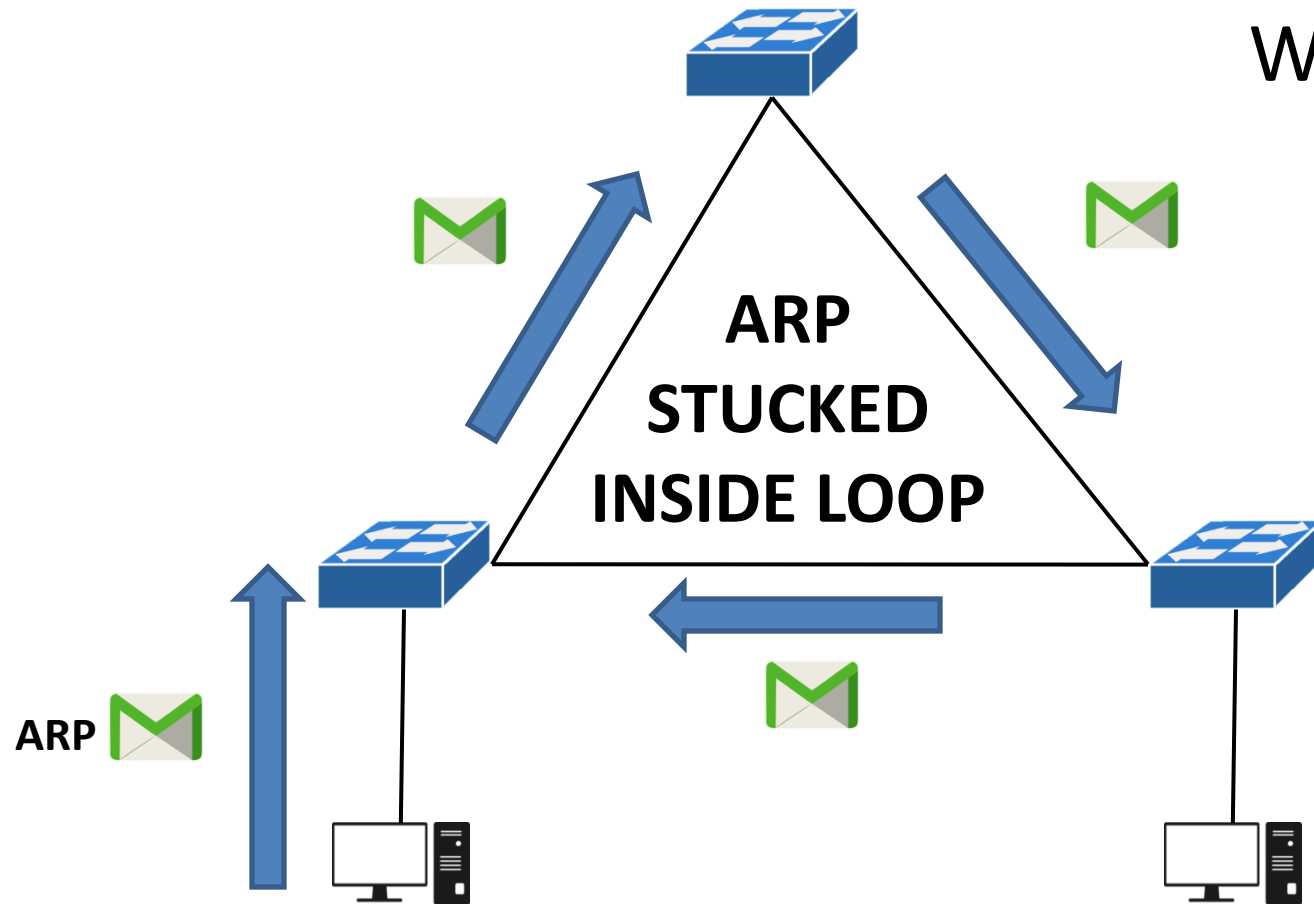
Spanning Tree Protocol (STP) Overview:

The Spanning Tree Protocol (STP) is a network protocol designed to prevent loops in Ethernet networks. Loops can cause significant issues such as broadcast storms and multiple frame copies, which can severely degrade network performance. STP was standardized as IEEE 802.1D.

Key Concepts of STP

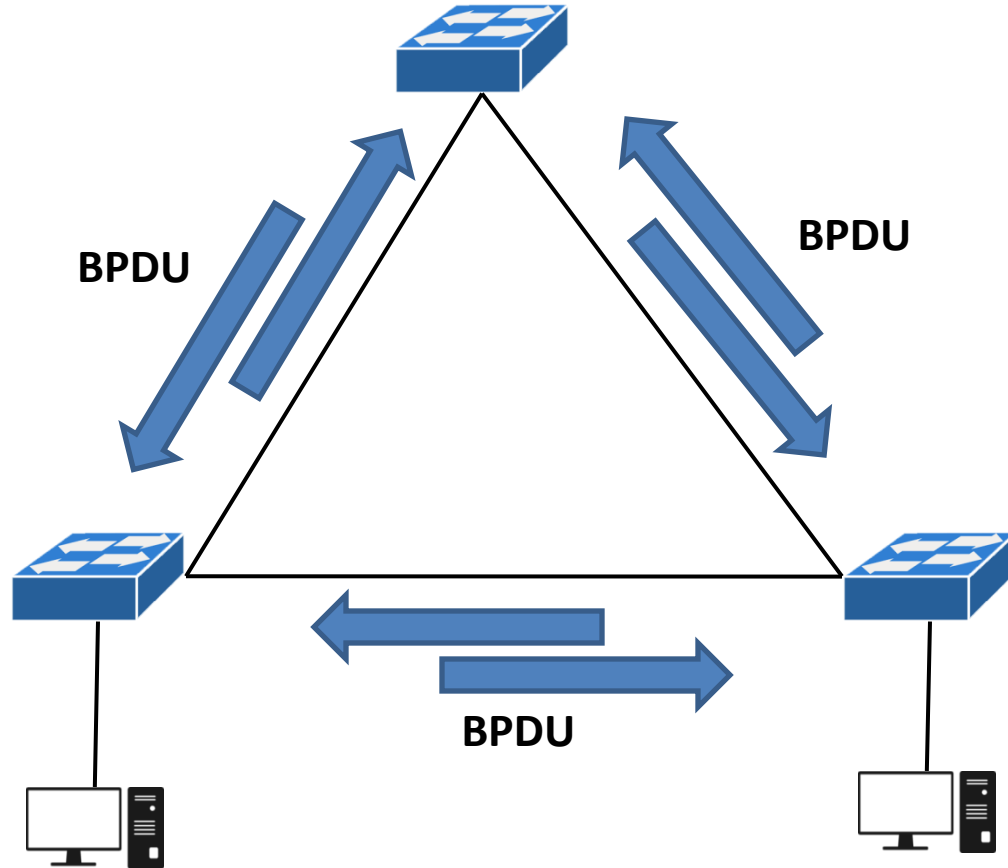
- 1. Bridge Protocol Data Units (BPDUs)**
- 2. Root Bridge**
- 3. Path Cost**
- 4. Port Roles**
- 5. Port States**

Without STP:



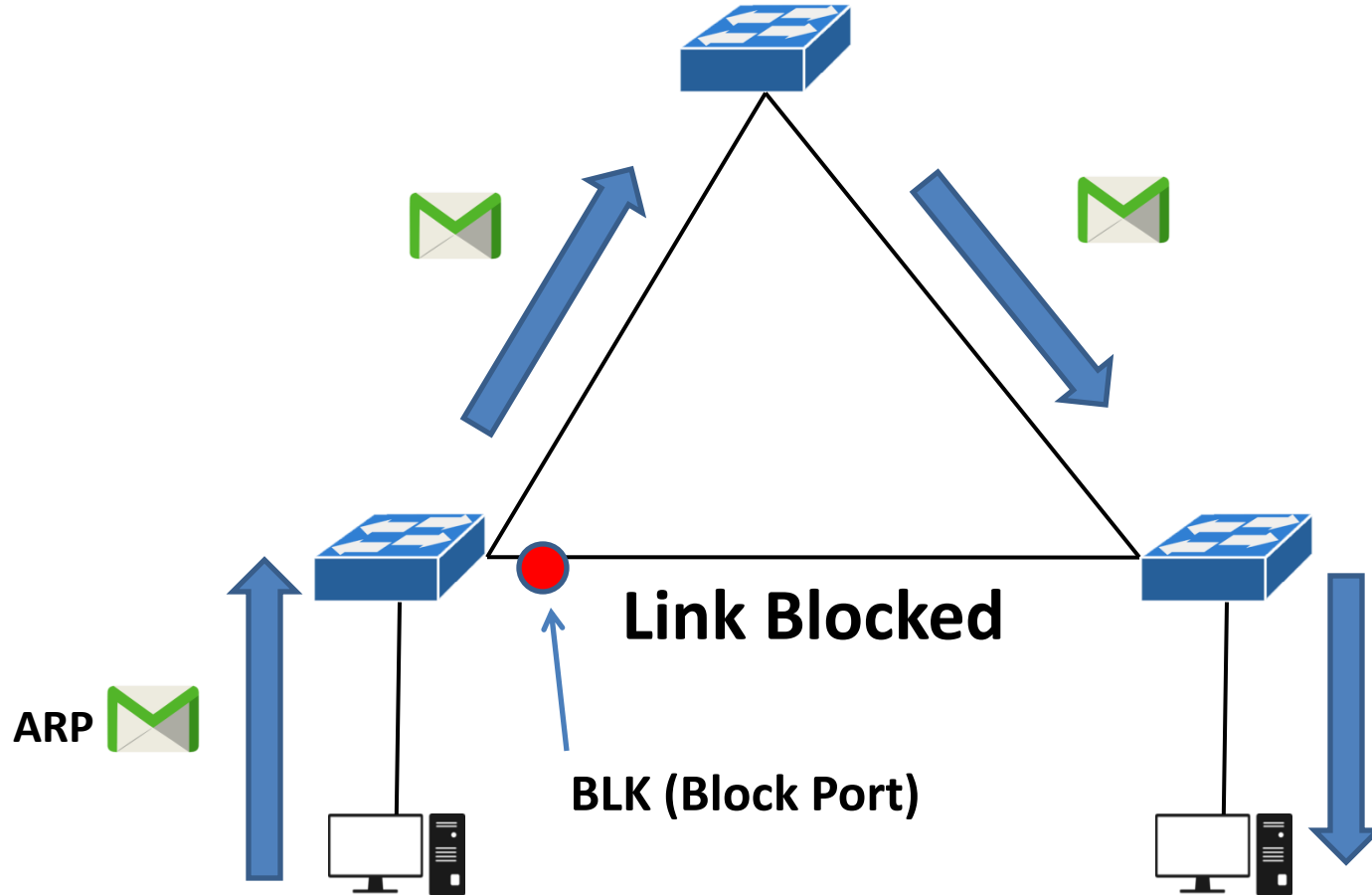
Without STP,
ARP
Stuck
Inside
Loop

When STP Working Switches Exchanges BPDU:



When STP working:

<https://www.youtube.com/@techwithravish>



BPDU:



DMA: Destination MAC address

SMA: Source MAC address

L/T: Frame length

LLC header: Logical link control header

Payload: BPDU data

Fields	Byte
Protocol ID	2
Protocol version ID	1
BPDU type	1
Flags	1
Root ID	8
Root path cost	4
Bridge ID	8
Port ID	2
Message age	2
Max age	2
Hello time	2
Forward delay	2

STP Root Bridge Election:

Switch with Lowest Bridge ID Will Become Root Bridge.

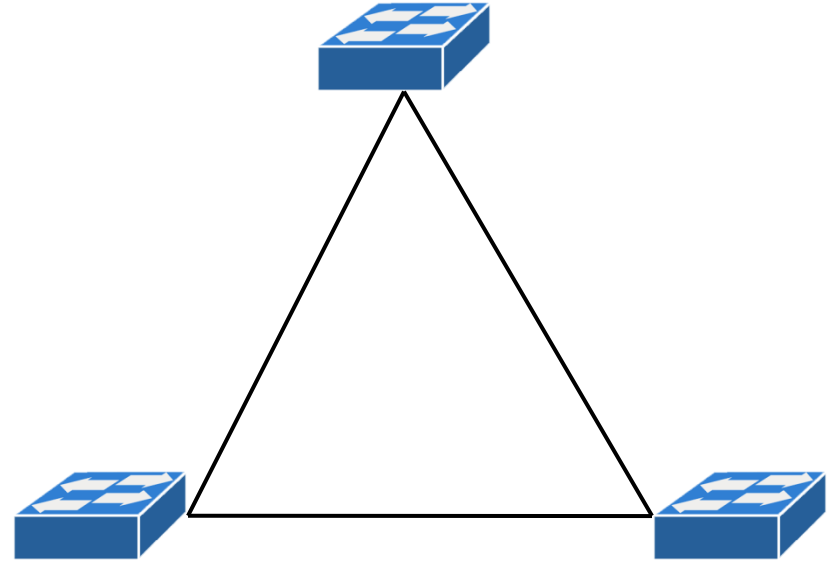
Rule 1:

Check Priority Number and the Switch with Least priority number become Root Bridge.

Rule 2:

If Priority Number is same then the lowest MAC Address will become the ROOT Bridge.

BID: Switch 0 (Root Bridge)
32768 +
0000.0C3C.68CA



BID: Switch 1
32768 +
0010.11C7.2DCB

BID: Switch 2
32768 +
000C.85E7.D356

Key Concepts of STP

Bridge Protocol Data Units (BPDUs):

- BPDUs are frames sent by switches to exchange information about the network topology.
- BPDUs are used to elect the root bridge and for the ongoing exchange of configuration messages.

Root Bridge:

The root bridge is the central reference point for all spanning tree calculations.

It is elected based on the lowest bridge ID, which consists of a priority value and the MAC address of the switch.

Key Concepts of STP

Path Cost:

The path cost is a value assigned to each port based on the speed of the link. The slower the link, the higher the cost.

STP calculates the path cost to determine the shortest path to the root bridge.

Link Speed	Path Cost	Range
16 Mbps	62	1-65535
100 Mbps	19	1-65535
1 Gbps	4	1-65535
10 Gbps	2	1-65535

Key Concepts of STP

Port States:

Blocking: Does not forward frames and does not learn MAC addresses.

Listening: Processes BPDUs but does not forward frames and does not learn MAC addresses.

Learning: Learns MAC addresses but does not forward frames.

Forwarding: Forwards frames and learns MAC addresses.

Disabled: Does not participate in frame forwarding or STP.

Key Concepts of STP

Port Roles:

Root Port (RP): The port on a non-root switch that has the lowest path cost to the root bridge.

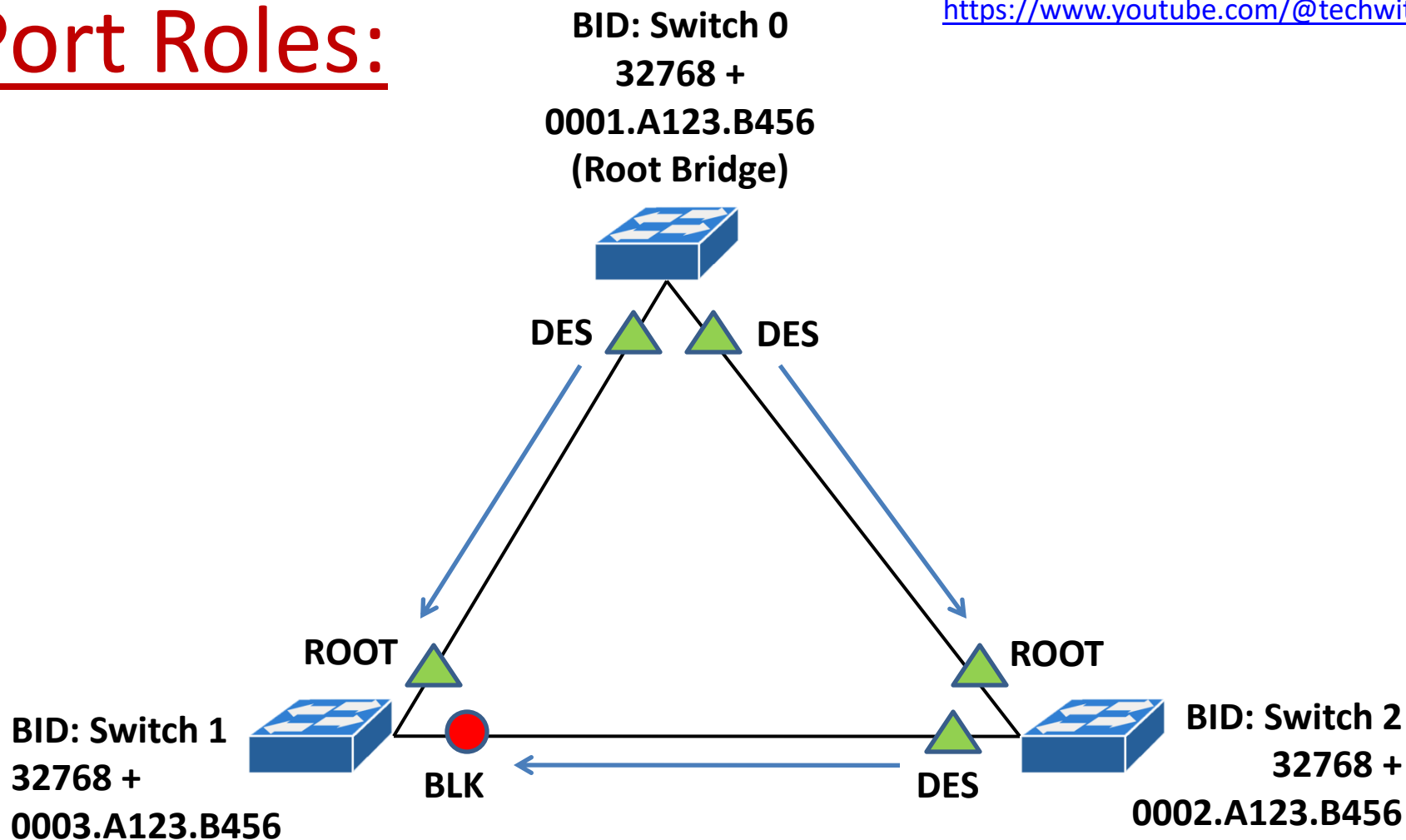
Designated Port (DP): The port on a network segment that has the lowest path cost to the root bridge.

Non-Designated Port: A port that is neither a root port nor a designated port. These ports are typically put into a blocking state to prevent loops.

Blocking Port: A port that does not forward frames to prevent loops.

Port Roles:

<https://www.youtube.com/@techwithravish>



STP Operation:

Root Bridge Election:

- All switches initially consider themselves as the root bridge.
- Switches exchange BPDUs to elect the root bridge based on the lowest bridge ID.

Path Cost Calculation:

- Once the root bridge is elected, each switch calculates the cost to reach the root bridge.
- The port with the lowest cost path to the root bridge becomes root port.

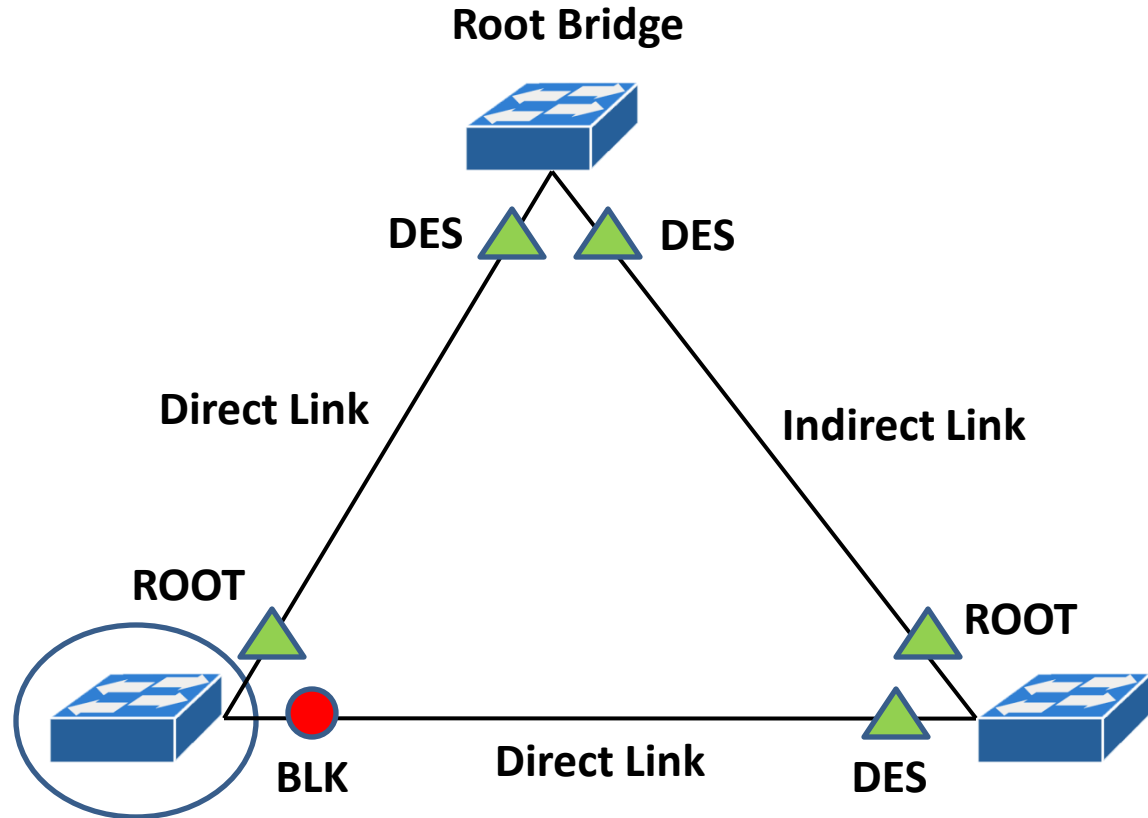
Port Role Assignment:

- Each segment will have one designated port that provides the shortest path to the root bridge.
- Non-designated ports are put into a blocking state to prevent loops.

STP Links:

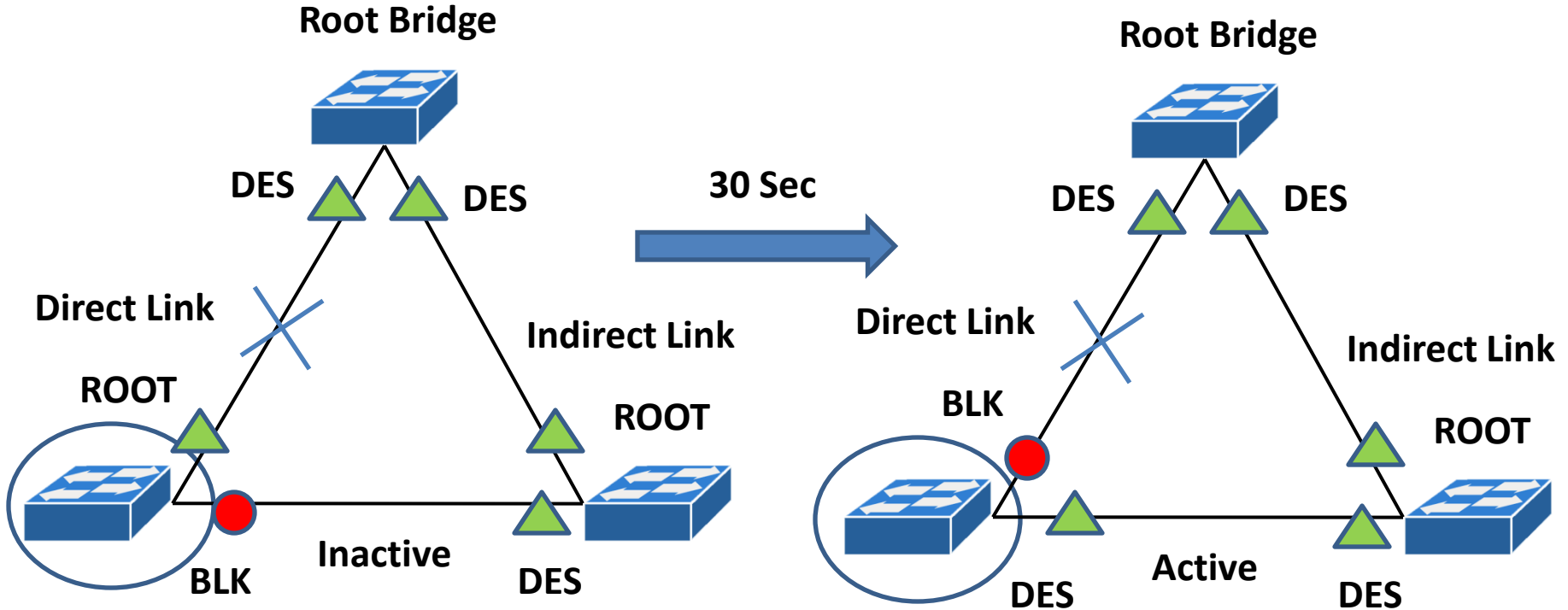
Direct Link: Link Adjacent or Directly Connected to Switch is called Direct Link.

Indirect Link: Link Not Adjacent or In Directly Connected to Switch is called Indirect Link.



STP Convergence Time:

If Direct Link is Down: 15 Sec (LSN) + 15 Sec (LRN)= 30 Sec



STP Convergence Time:

If Indirect Link Down: 15 Sec (LSN) + 15 Sec (LRN)+ Wait 20 Sec= 50 Sec

